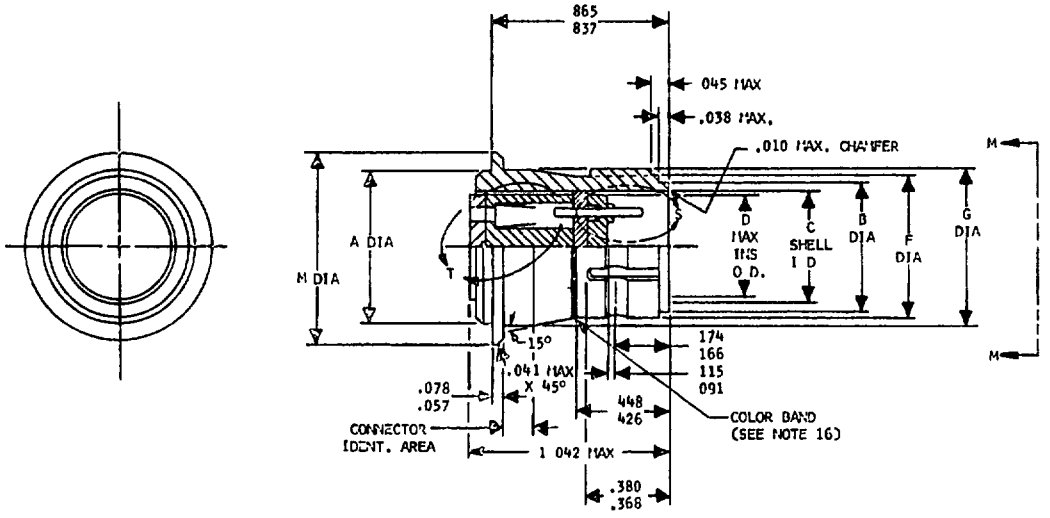
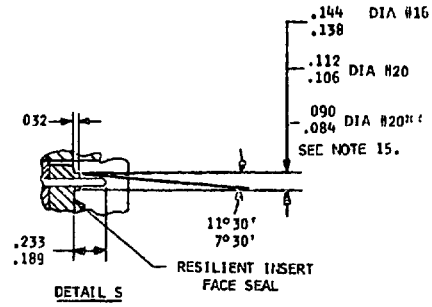
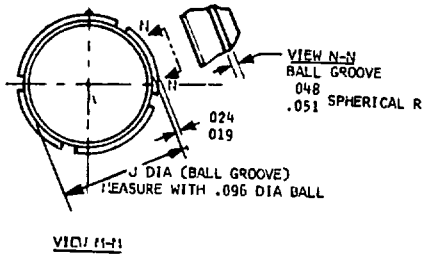
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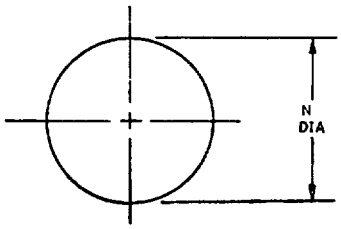


DETAIL S



VIEW N-N

FIGURE 1
CONNECTOR VIEW



PANEL CUTOUT

PART NUMBER EXAMPLE

MS3143 N 3 - 50 P Y
- Y: ALTERNATE INSERT POSITION
- P: CONTACT STYLE (P-PIN)
- 50: INSERT ARRANGEMENT
- 3: SHELL SIZE
- N: CLASS
- MS3143: MS NUMBER

[1] SEE SHEET 2 FOR DETAIL T

[2] ON OUTER ROW CONTACTS OF INSERT ARRANGEMENTS INDICATED IN CHART

INSERT ARRANGEMENTS REQUIRING REDUCED DIAMETERS FOR PIN CONTACT RAISED SEAL BARRIERS		
SHELL SIZE	INSERT ARRANGEMENTS	CONTACT CAVITIES
12	12-50	5, 6, 8, 9, 11 & 12

Review activities:
Navy-EC

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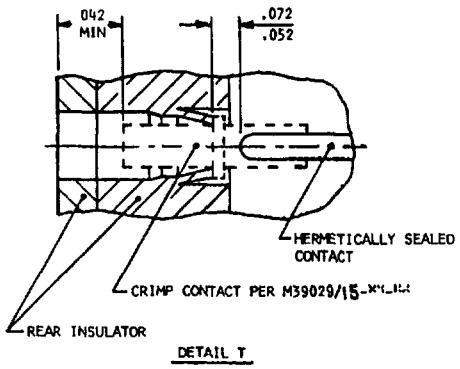
P A Navy-AS Other Cust	TITLE CONNECTOR, RECEPTACLE, ELECTRIC, HERMETIC, CRIMP TYPE, SOLDER MOUNTING, PUSH-PULL COUPLING, SERIES 3 (CLASS N)	MILITARY STANDARD MS3146(NAVY)
PROCUREMENT SPECIFICATION MIL-C-81703	SUPERSEDES	SHEET 1 OF 3

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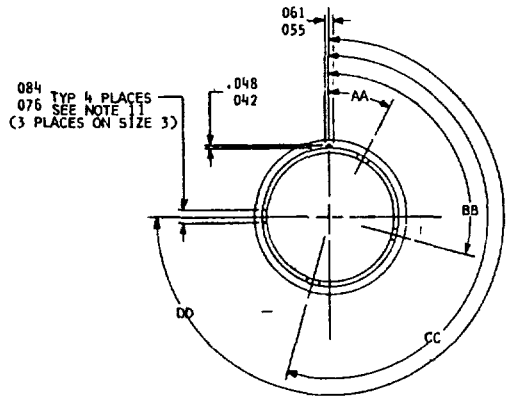


FIGURE II
KEYWAY POSITIONS

ENGAGING ENDS OF CONTACTS

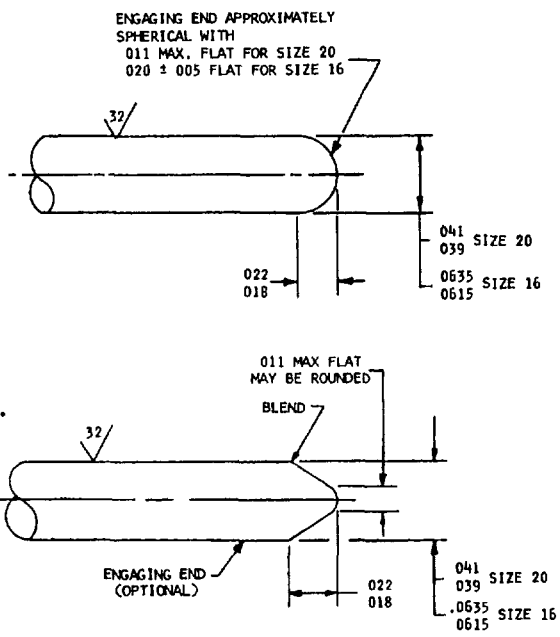


FIGURE III
CONTACT DETAILS

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TABLE I
CONNECTOR DIMENSIONS AND WEIGHTS

SHELL SIZE	A	B	C	D	E	G	J	M
3	.497	.392 .385	.367 .356	330	441 431	.531	398 .391	602 582
7	.622	.527 .520	.483 .473	451	.576 566	.671	533 .526	.760 .740
12	.747	.661 .654	.624 .614	553	.710 .700	812	.668 .661	885 .865
19	.904	.800 .793	.763 .753	.738	.849 839	.937	.806 .799	1.010 990
27	1.060	.955 .948	918 .908	854	1 004 .994	1 125	.961 .954	1 192 1.172
37	1 153	1.077 1 070	1 040 1.030	976	1 126 1 116	1 218	1 083 1.076	1 291 1 271

SHELL SIZE	N	WEIGHT (LBS) SEE NOTE 7 PIN INSERT
3	.515	(WEIGHT WILL BE
7	.640	SUPPLIED IN
12	.764	LATER REVISION)
19	.922	
27	1.077	
37	1.171	

TABLE II
KEYWAY POSITIONS

SHELL SIZE	AA	BB	CC	DD
3	23°	97°	187°	--
7	23°	97°	187°	277°
12	30°	105°	195°	270°
19	30°	105°	195°	270°
27	30°	105°	195°	270°
37	30°	105°	195°	270°

NOTES (NOTES 1 THRU 6 APPLICABLE TO CONTACTS ONLY)

1 CONTACT FINISH GOLD PLATE 000050 INCH MIN IN ACCORDANCE WITH MIL-G-45204. A SUITABLE UNDERPLATING MAY BE USED SILVER UNDERPLATING SHALL NOT BE USED FINISH IS APPLIED AFTER SEALING
2 DIMENSIONS IN INCHES, DIMENSIONS APPLY AFTER PLATING.
3 63 ALL OVER PER MIL-STD-10, UNLESS OTHERWISE SPECIFIED
4 REMOVE ALL BURRS AND SHARP EDGES 003 TO .005 R.
5 DIAMETERS TO BE CONCENTRIC WITHIN 004 TIR
6 DIAMETERS IN THE SAME CROSS-SECTIONAL PLANE TO BE CONCENTRIC WITH EACH OTHER WITHIN .003 TIR.
7. WEIGHT LISTED IS CONNECTOR MAXIMUM WEIGHT PER SHELL SIZE LESS CRIMP CONTACTS
8 SHELL MATERIAL COLD ROLLED STEEL, PER QQ-S-637
9 SHELL FINISH 0001 MIN TIN PER MIL-T-10727, OVER NICKEL PER QQ-N-290
10 EACH CONNECTOR SHALL BE PACKAGED WITH A PLASTIC DUST CAP OVER EACH END OF SHELL
11 FOUR KEYWAYS (MMC) AND INSERT SHALL BE LOCATED WITHIN .004 EITHER SIDE OF (TP) RELATIVE TO MASTER KEYWAY (MMC) AND SHELL O.D (MMC)
12 DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCE SHALL BE XX ± .010 AND .XXX ± .005, ANGULAR TOLERANCE, X° ±1°-0' AND X°X' ± 30'
13 INSERTION/REMOVAL TOOL, MS3160-[illegible] TO BE USED WITH THIS CONNECTOR
14 CONTACT MIL-C-39029/15-[illegible] MUST BE USED FOR WIRE TERMINATIONS.
15 DIAMETERS SHOWN ARE AT BASE OF BARRIER.
16 COLOR BAND (A) 070 ± 031 WIDE, COLOR BLUE
(B) BAND LOCATION MUST BE SUCH THAT IT IS VISIBLE WHEN MOUNTED
17 MAXIMUM WIRE DIAMETER PERMISSIBLE 074 FOR SIZE 20 CONTACT, .103 FOR SIZE 16 CONTACT

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PROCUREMENT SPECIFICATION MIL-C-81703	SUPERSEDES	SHEET 3

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